02/12

Wednesday, 12 February 2025 9:39 AM

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$



When does 2nd order derivative represent

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \mu\frac{\partial^2 u}{\partial x^2}$$

Even order → diffusion

odd order → derivatives } amplitude only depends on the

$$\frac{u_i^{n+1} - u_i^n}{\Delta x} + c\frac{u^{n+1} - u^n}{\Delta x} = \mu\left(\frac{u^{n+1} - u^n + u^{n-1}}{(\Delta x)^2}\right)$$

Shortest wavelength is the issue ...

Modified equation method

Courant number & CFL condition

↓

$= 1$

(one grid space in 1 time step)

↳ domain of influence of FD and ODE

Non-linear problem :

$$\frac{\partial T}{\partial t} = \alpha' \frac{\partial^2 T^{(m)}}{\partial x^2} \qquad m > 1$$

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha' \left(T_{i+1}^n\right)^{(m)} - 2\left(T_i^n\right)^{(m)}$$

$$F\left(T_{i\pm1}^n\right) = F\left(T_i^n\right) \pm \frac{\partial F\left(T_i^n\right)}{\partial x}\Delta x + O(\Delta x^2)$$ } First order approx.

↓

$$\frac{\partial F}{\partial T}\frac{\partial T}{\partial x}\Delta x \rightarrow \frac{\partial F}{\partial T}\left(\frac{T_{i+1} - T_i}{\Delta} + O(\Delta x)\right)\Delta x$$

$$\frac{\partial F}{\partial T}\frac{\partial T}{\partial x}\Delta x \rightarrow \frac{\partial F}{\partial T}\left(\frac{T_{i+1}-T_i}{\Delta x}+O(\Delta x)\right)\Delta x$$

$$=\left(\frac{\partial F}{\partial T}\right)(T_{i+1}-T_i)$$

$$\uparrow$$

$$F = T^{(m)}$$